

Algebra II with Trigonometry

Chapter 7.1

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:
[4564659_alg-2trig-h-chp-7-1-note.pdf](#)

Problems

1. Without using a calculator, compute $\frac{\sin 20^\circ + \sin 40^\circ + \sin 80^\circ}{\cos 20^\circ + \cos 40^\circ + \cos 80^\circ}$.
2. Simplify $\frac{1 - \tan^2 x}{1 + \tan^2 x}$ without converting to sines and cosines, and find its value when $x = 22.5^\circ$.
3. If $\tan \theta + \cot \theta = 4$, compute the possible values of $\sin 2\theta$.
4. If $\sin \alpha + \sin \beta = \sqrt{2}$ and $\cos \alpha + \cos \beta = \sqrt{2}$, find the exact value of $\sin(\alpha + \beta)$.
5. Evaluate $\frac{\sec A - 1}{\sec A + 1} + \frac{\sec A + 1}{\sec A - 1}$ in simplest form for $0 < A < \frac{\pi}{2}$.
6. Let $x = 3 \sec \theta$ and $0 < \theta < \frac{\pi}{2}$. What is the value of $\sqrt{x^2 - 9} - \sqrt{x^2 - 3x}$ in terms of θ ?